QC STORY PROJECT

PROJECT TITLE :- To eliminate the concern of High Mount Stop Lamp poor fit from Marazzo model.

Mahindra and Mahindra Ltd, Nasik Plant, Scorpio TCF.

	Plan	Actual
Start Date	01.08.18	15.08.18
Completion Date	30.11.18	20.12.18

Current Status:

closed

Phase II – QC Story

- Team Information
- Step1: Select the Theme
- Step 2: Justify the choice
- Step 3: Understand the current situation
- Step 4: Select Targets
- Step 5: Analysis (Level 1 Analysis)
- Step 6: Implement corrective measures
- Step 7: Confirm the Effects
- Step 8: Standardize
- Step 9: Summarize & Plan future actions
- Guidelines to user

2. Justify the choice (reasons)

Mahindra

PROMISE 2019
Automotive Business

We will...

Double our revenue to ₹53,000 cr (USD 8 bn), while delivering superior financial performance

- Be the 2nd largest OEM in India by volume
- Double the overseas volume with significant presence in 10 markets

By being the most customer centric company

Brand and Customer Satisfaction
- Be among the Top 2 in Customer Satisfaction
- Be among the Top 3 in brand ranking in PV and #1 in UV & SCV
- Have 10 brands with CaPS > 50

Product and Technology
- Launch 18 products
- Be a pioneer in digital customer experience
- Develop one new disruptive business model

Crusade 2.0
- Drive brand preference through Crusade 2.0 of delivering ETR, FTR and DCE

People
- Be the #1 Great Place to Work in the manufacturing sector

Environment
- Reduce carbon and water footprint by 25%

Thus enabling our customers and stakeholders to Rise, as we progress towards being a Top 10 Global Automotive brand.

Mahindra

PROMISE 2019
Nashik Plant 1, Auto Division

We will...

- Produce 230k vehicles in a year as per market requirement
- Fulfill 100% IO requirement for doubling the overseas volume
- Maintain cost leadership in Auto Sector by reducing revenue cost of 5% YoY
- Improve IQS for Nashik models by 25%, ranking in top 3
- Achieve CaPS > 50 for all Nashik models
- Launch 7 new products on time
- Achieve defined Crusade targets through FTR, ETR and DCE
- Achieve operational excelle[illegible] through digitization
- Achieve No.1 status for em[illegible]ee engagement in the Auto Industry
- Make officers future ready [illegible]plement new technologies coming in the next 5-10 ye[illegible]
- Reduce carbon & water foo[illegible]nt by 25%

Departmental Objective –

DPR Improvement

Cross Functional Team

No.	Description	Name	Department	Mobile Number
1	Team Leader	Pravin Patil	Tool Room	8888897850
2	Mentor	Rahul Narkar	Production	9881080299
3	Member 3	Sakharam Shelke	Production	9881900427
3	Member 3	Sachin Bhambre	MQA	9158960111
4	Member 4	Ganesh Bendale	Production	9850952219
5	Member 5	Abhishekh Sonawane	Production	9881751802

1. Select the Theme

Brief description of Problem

- What : High Mount Stop Lamp poor fitted
- Where : At trim line - 3
- When : From start of May'18.
- Who : High Mount Stop Lamp fitment associate
- Why : Repeat concern at buyoff, FAI and CAI
- How : Average 85 concerns per month.

HSML poor fitted

2. Justify the choice (reasons)

Problem Criticality (Severity)

- **Importance & Severity of Concern :-**

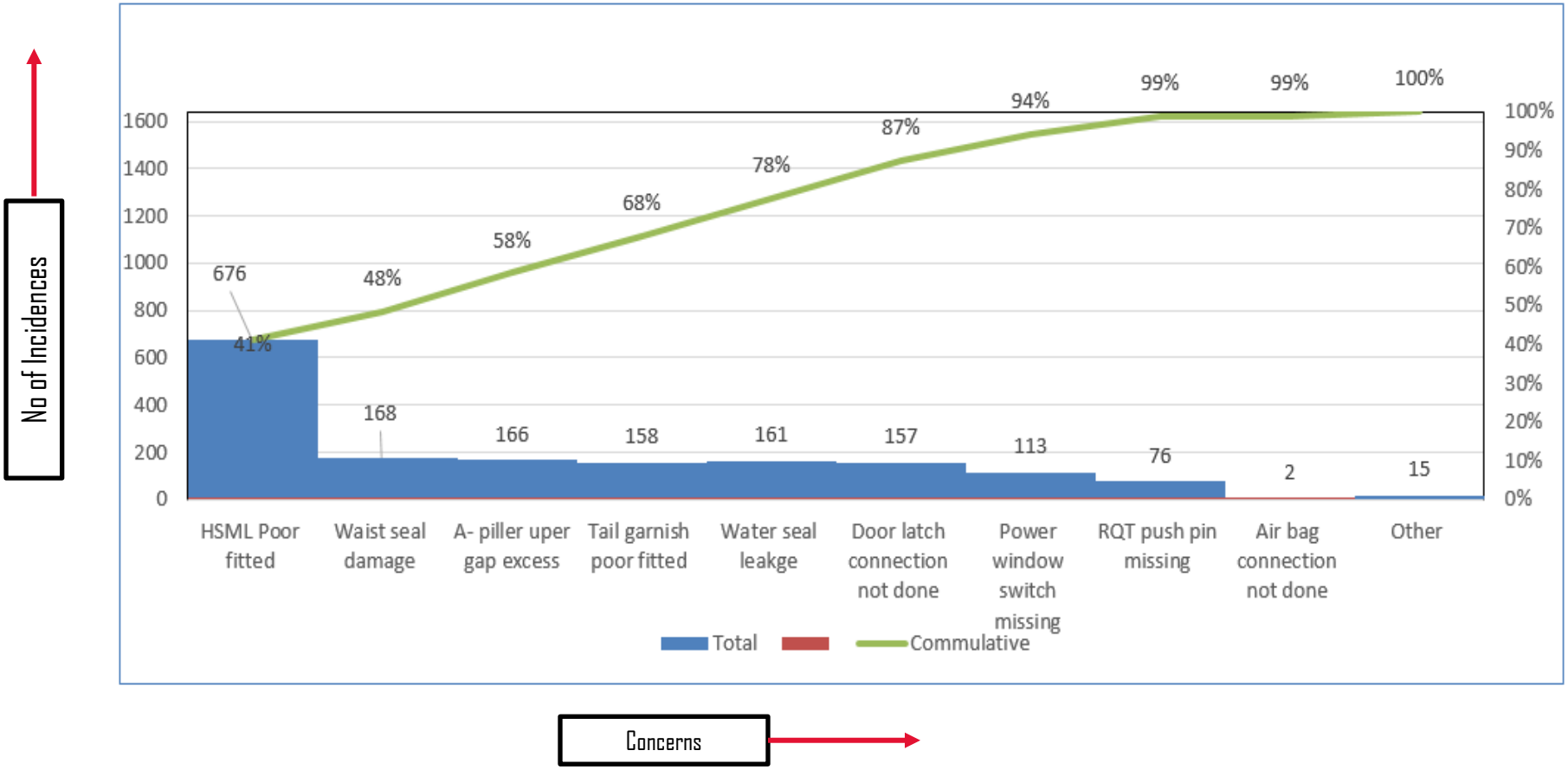
- High Mount Stop Lamp **poor fitted concern is V1 concern.**
- **Major contributor in productivity & FRC loss concern.**
- **Skill operator required for rework.**
- **Rework activity is critical.**
- **Repair cost is very high i.e. -------per year.**

Rework Cost Calculations :-

Rework Cost estimation		
Description	Units	
Rework time (Approx)	Min	10
Avg Concerns/Day	Nos	6.0
Total Required Rework time per day	Min	60
Effective working time	Min	465
Operator required /day	nos	0.13
Approx Operator cost/ day	Rs	1360
Operator cost /concern	Rs	175.48
Avg no. of working days	days	25
Operator cost /month	Rs	4387.1
Operator Cost/year	**Rs**	**52645.2**

2. Justify the choice (reasons)

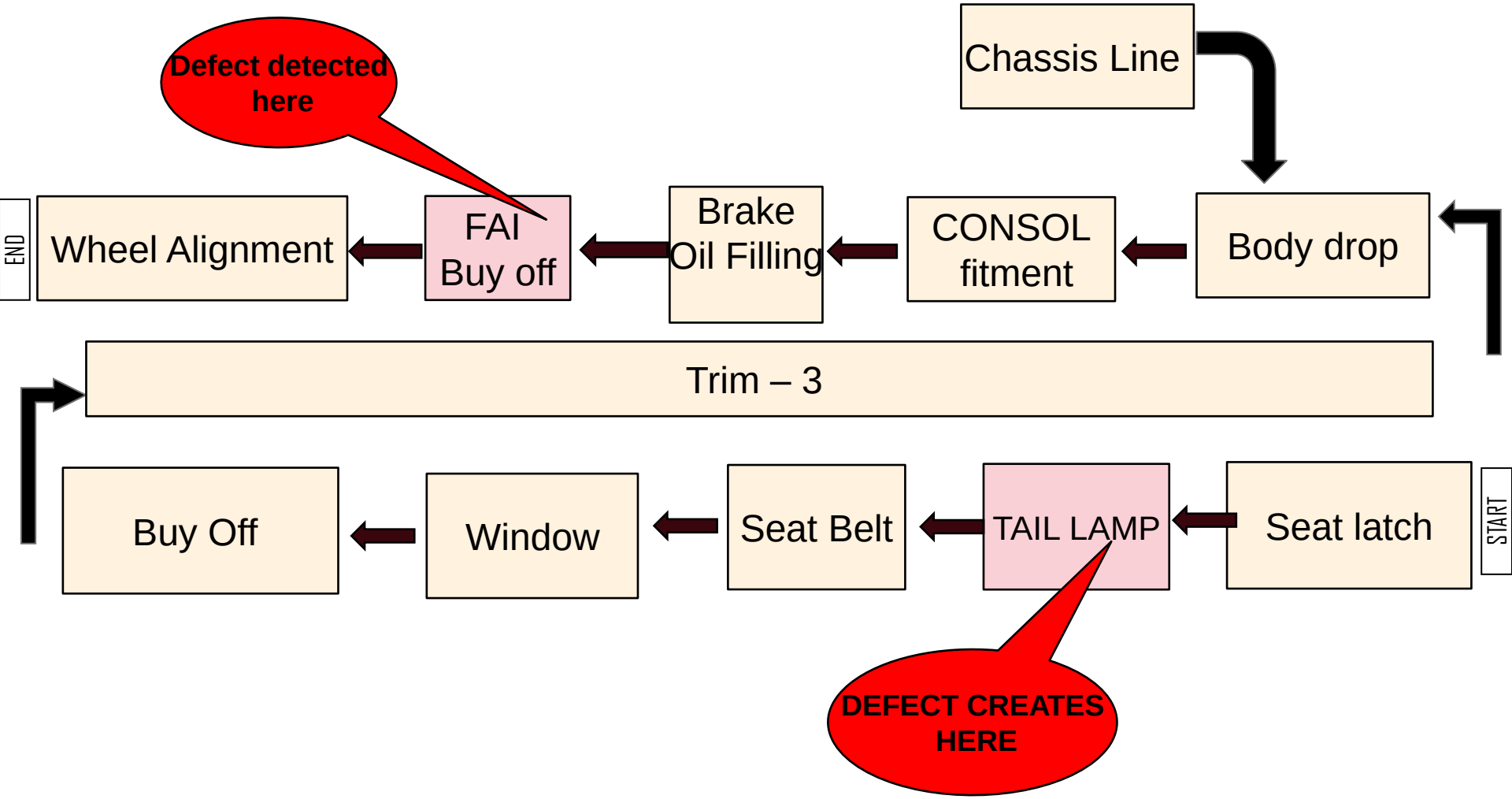
Pareto of May to Sep'18



Inference – After Pareto analysis it is observed that HMSL poor fit is top concerns in QDMS Buy off

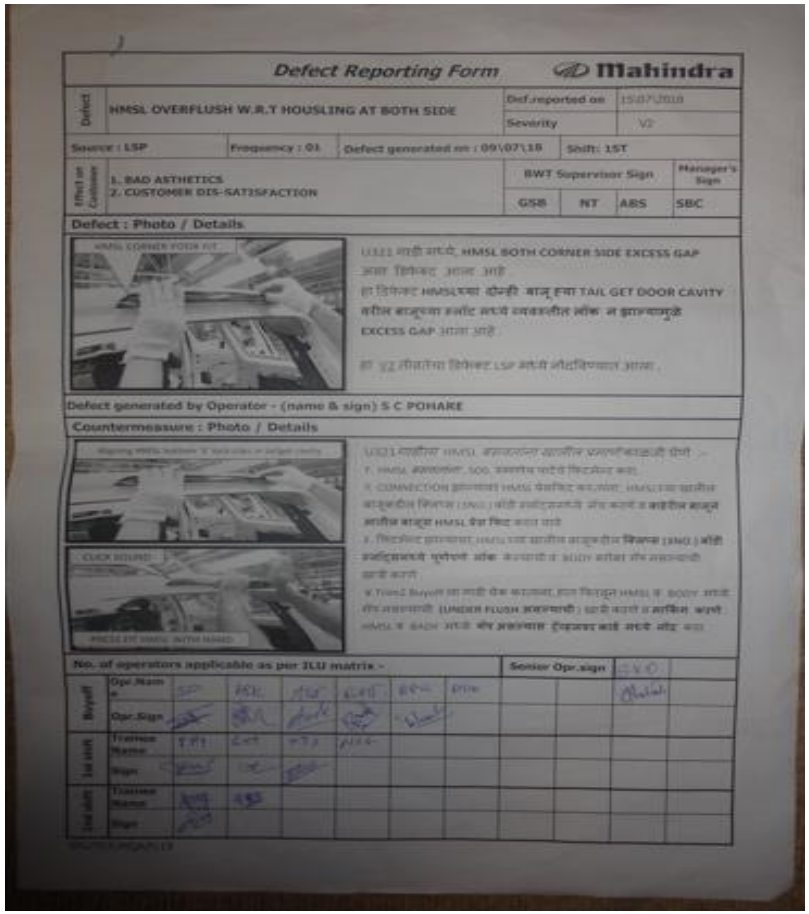
3. Understand the current situation

Observation of process



3.a Immediate Containment Action

- DRF & OPL given to operators for Concern detection at stage & buyoff.



Defect Reporting Form — Mahindra

Defect	HMSL OVERFLUSH W.R.T HOUSLING AT BOTH SIDE	Def.reported on	15\07\2018
		Severity	V2
Source : LSP	Frequency : 01	Defect generated on : 09\07\18	Shift: 1ST
Effect on Customer	1. BAD ASTHETICS 2. CUSTOMER DIS-SATISFACTION	BWT Supervisor Sign	Manager's Sign
		GSB / NT / ABS	SBC

Defect : Photo / Details

HMSL CORNER FOUR FIT

U321 गाडी मध्ये, HMSL BOTH CORNER SIDE EXCESS GAP आला डिफेक्ट आला आहे

हा डिफेक्ट HMSLच्या दोन्ही बाजू हया TAIL GET DOOR CAVITY वरील बाजूच्या ब्लॉट मध्ये व्यवस्थीत लॉक न झाल्यामुळे EXCESS GAP आला आहे.

हा 32 नंबरचा डिफेक्ट LSP मध्ये नोंदविण्यात आला.

Defect generated by Operator - (name & sign) S C POHARE

Countermeasure : Photo / Details

CLICK SOUND

PRESS FIT HMSL WITH HAND

U321 गाडीला HMSL बसवताना खालील प्रमाणे कार्यवाही घेतली :-

१. HMSL बसवणेपूर्वी, SOD. प्रमाणे गाडीचे फिटमेंट करा.

२. CONNECTION झाल्यानंतर HMSL फेसिंग करताना, HMSL च्या खालील बाजूकडील क्लिपच (JIG) बोटे दाबूनच बसवणे व वरीलप्रमाणे बाजूने खालील बाजूस HMSL दैव फिट करा व

३. फिटमेंट झाल्यानंतर HMSL च्या खालील बाजूकडील क्लिपच (JIG) बोटे दाबूनच पूर्णपणे लॉक केल्याची व BODY बरोबर सेट असल्याची खात्री करणे.

४. Trim2 Buyoff ला गाडी घेऊन जाताना, हात फिरवून HMSL व BODY मध्ये सेट असल्याची (UNDER FLUSH असल्याची) खात्री करणे व आवश्यक असल्यास HMSL व BODY मध्ये सेट असल्याशिवाय ट्रॉलीवर काढी नंतरचे नोंद करा

No. of operators applicable as per ILU matrix - | Senior Opr.sign

Buyoff	Opr.Name									
	Opr.Sign									
1st shift	Trainee Name									
	Sign									
2nd shift	Trainee Name									
	Sign									

Defect communication done through DRF to all stage associate. & Buy off associate.

3. Understand the current situation

Observation of HMSL fitment on tail lamp stage

HMSL Pick up from hardware trolley

Position HMSL on body

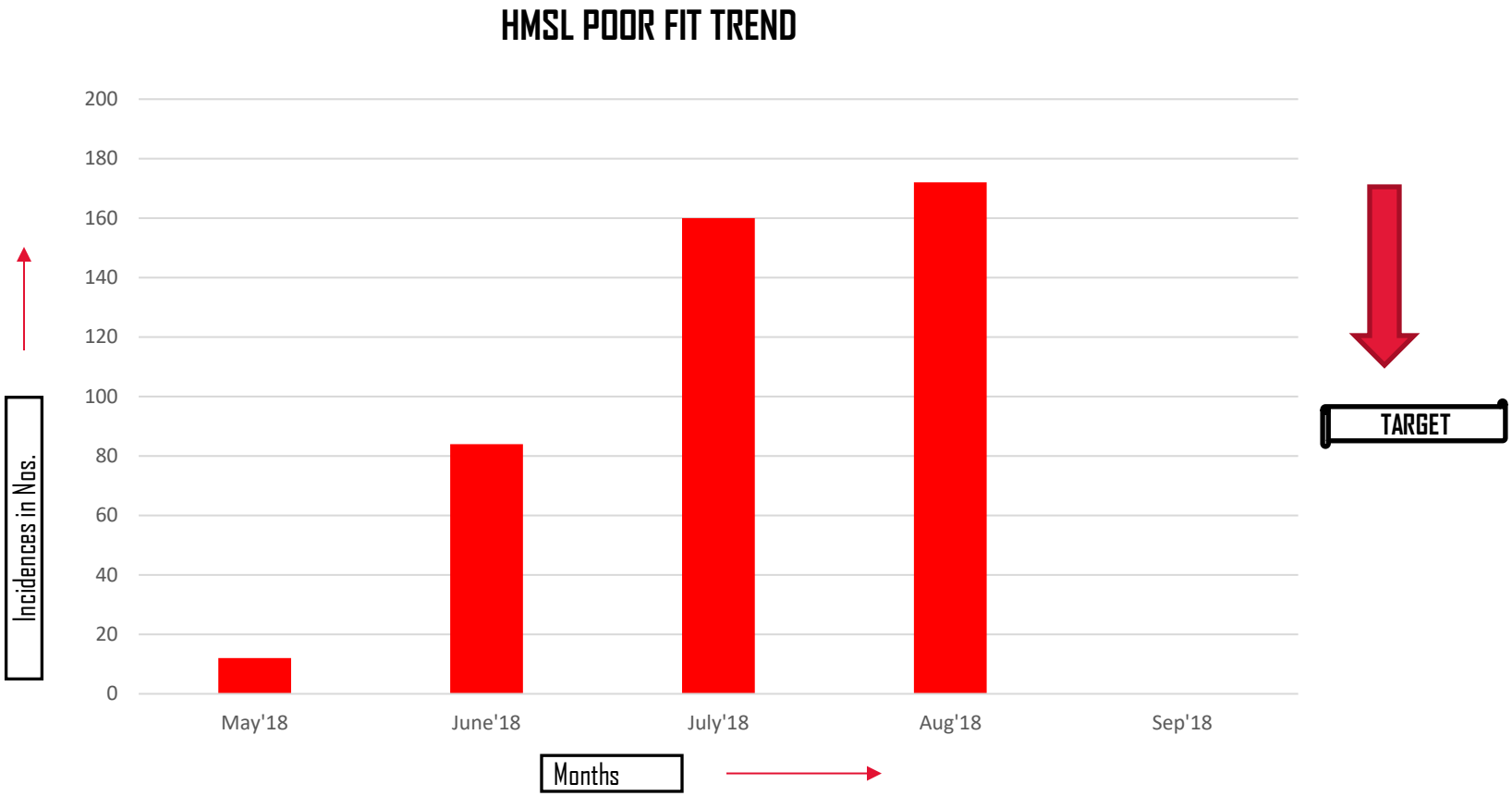
Position HMSL on body

Press pin and fit HMSL on body

Align and match lock pin

Insert wire pigtail

3. Understand the current situation

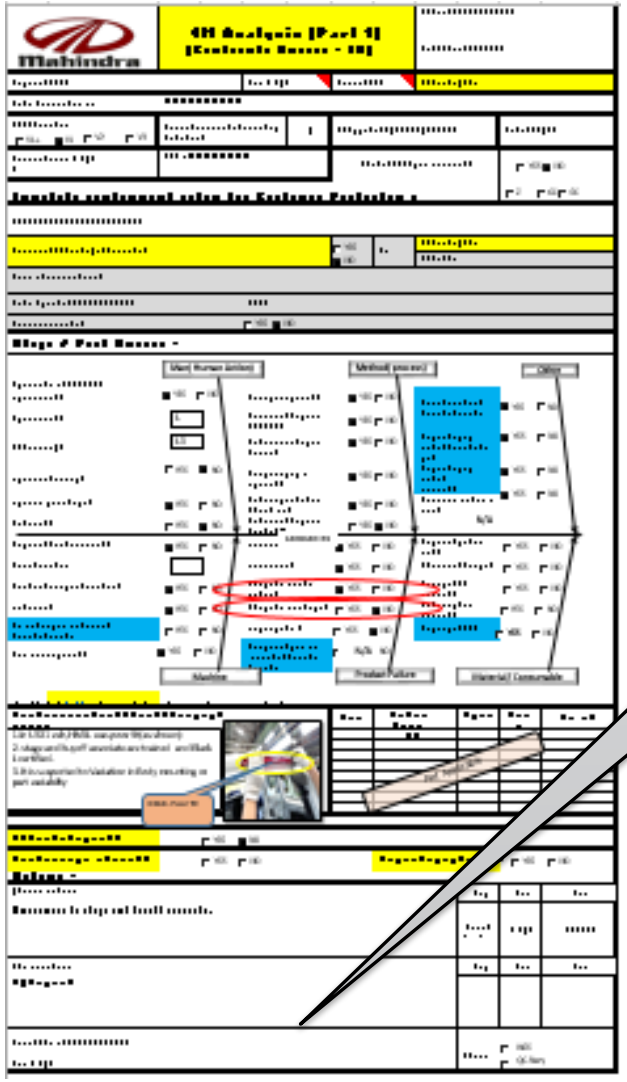


DPT of Vaccum leak test Is not ok (May to Sep'18) = R/1000.

Inference – Average 85 concern per month reported at Trim line Buy off

3. Understand the current situation

L0 for HMSL poor fitted concern in Marazzo model.



- Inference from L0
Problem follows part
- Problem is a repeat problem
L1 analysis required.

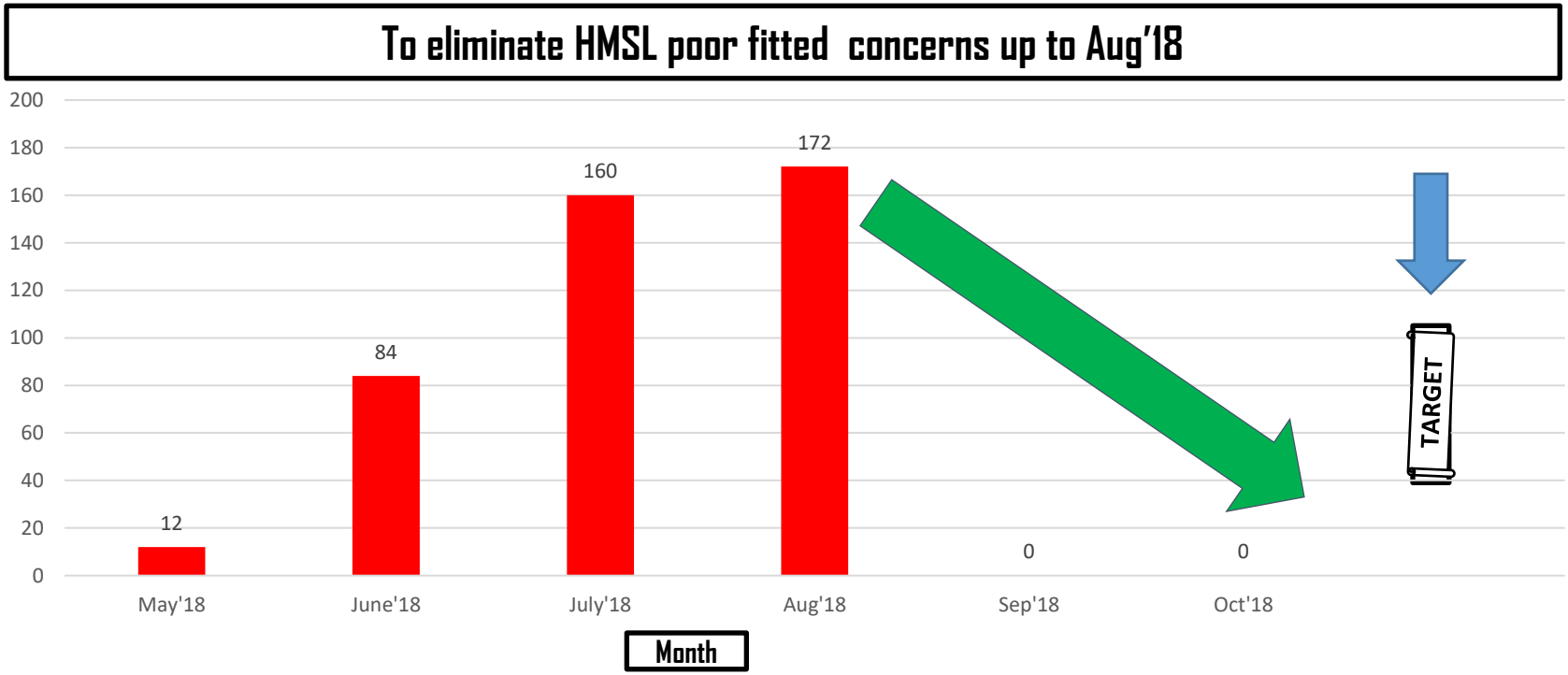
4M analysis Good and Not so good observations	
MAN	Operator dext. Level L3/L4
Method	HMSL fitment on body
Others	Control plan adhered on stage
Machine	N/A
Product failure	
Material	Material stored as per requirement.

V1

4. Select Target

Identify the target. A target is a number indicating the level of improvement to be achieved.

Present Level:	Average 85 vehicles reported for m leak test in a month
Target:	To eliminate the concern of HMSL poor issue from Marazzo model by Oct-18



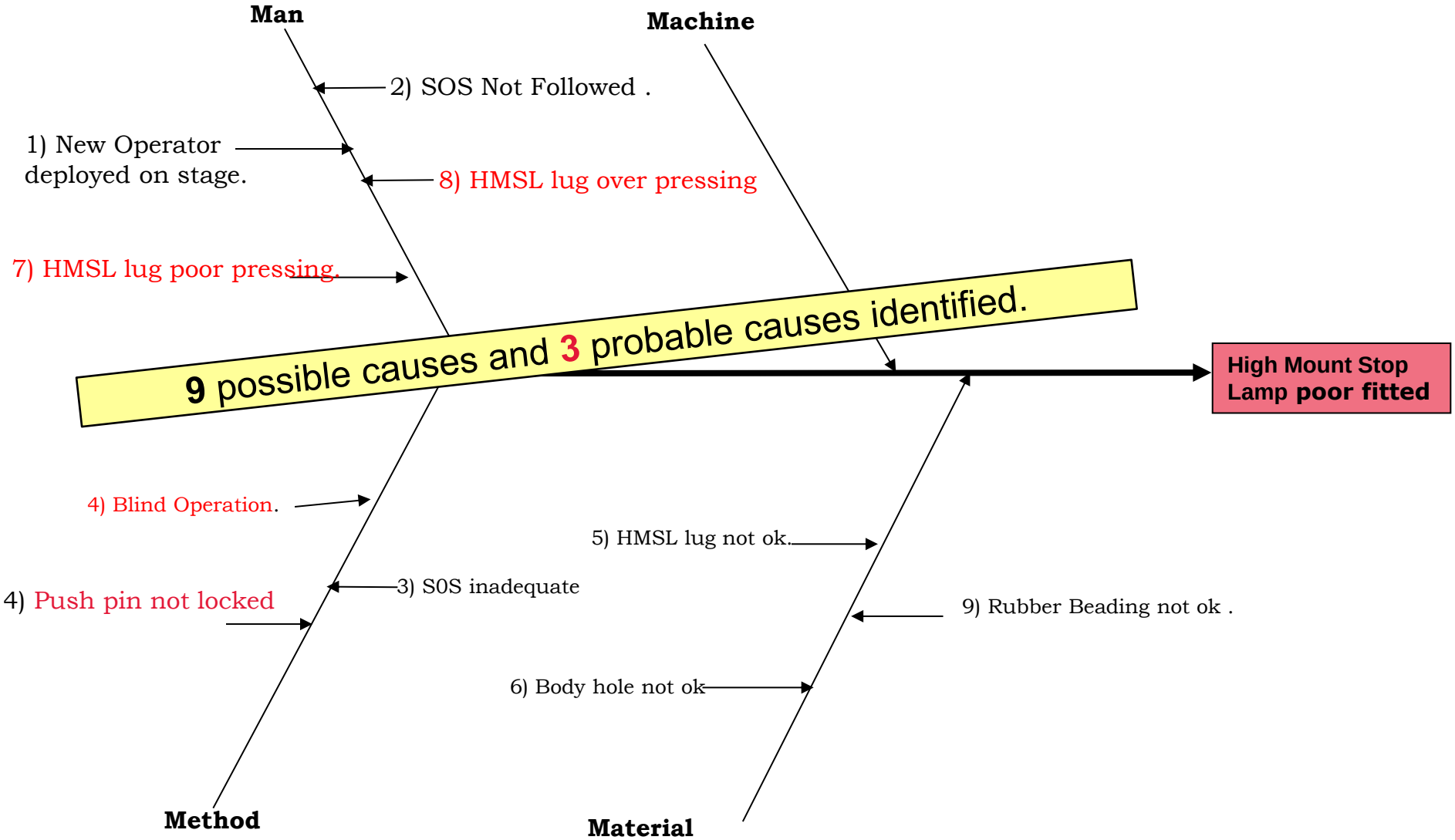
5. Analysis

List of Possible causes after brainstorming among team members

Sr. no	Possible Causes
1	New Operator deployed on stage
2	SOS Not Followed
3	SOS inadequate
4	Push pin not locked
5	High Mount Stop Lamp lug not ok.
6	body hole not ok
7	High Mount Stop Lamp lug poor pressing
8	High Mount Stop Lamp lug over pressing
9	Rubber beading not ok

5. Analysis

5.1 Identify possible causes: Fish Bone diagram.



5. Analysis

5.1 List of possible causes

Sr. no	Possible Causes	Pravin Patil	Rahul Narkar	Sakharam Shelke	Sachin Bhambre	Ganesh Bendale	Abhishekh Sonawane	Total
1	New Operator deployed on stage.	3	1	3	3	3	3	16
2	SOS Not Followed .	9	9	0	0	1	3	22
3	SOS Inadequate	0	0	1	1	3	3	8
4	Push pin not locked	9	9	1	3	9	9	40
5	High Mount Stop Lamp lug not ok.	9	0	1	1	3	9	23
6	Body hole not ok	3	9	9	9	9	3	42
7	High Mount Stop Lamp lug poor pressing.	9	9	9	3	9	9	48
8	High Mount Stop Lamp lug over pressing.	9	9	9	3	9	3	42
9	Rubber beading not ok	9	0	1	1	3	3	17

Scale: 0=None, 1=Low, 3=Moderate, 9=Strong
Threshold Limit = 39

Step 5. Analysis

List of causes not considered as probable causes and reasons for not considering

Sr. no	Not Possible Causes	Reasons
4	New Operator deployed on stage.	All trained operator deployed on stage
2	SOS Not Followed .	SOS followed
3	SOS Inadequate	SOS as per POS ref.
5	High Mount Stop Lamp lug not ok.	Part checked found ok as per drwg.
9	Rubber beading not ok	Rubber beading found ok

5.1 TEST OF HYPOTHESIS

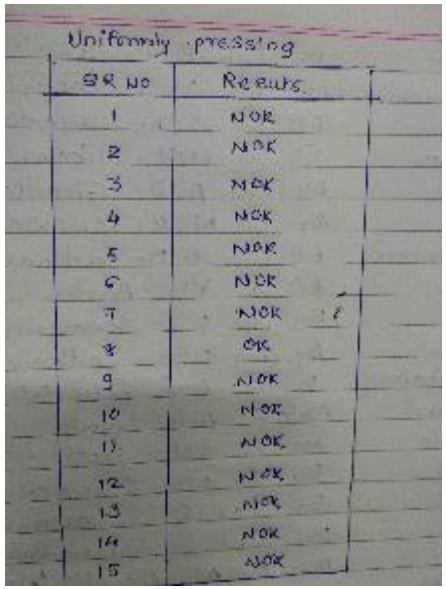
Confirm the probable cause:-

Sr. no	Probable Cause	Reasons
4	Push pin not locked	No reference to installation of High Mount Stop Lamp on body
6	Body hole not ok	High Mount Stop Lamp fitment hard due to hole mismatch with lug
7	High Mount Stop Lamp lug poor pressing.	Pressing of High Mount Stop Lamp on body press by hand
8	High Mount Stop Lamp lug over pressing.	Lug press by tapping on High Mount Stop Lamp

5.3 Confirm the probable cause:

No	Probable Cause	Testing method	Observations	Conclusion
4	Push pin not locked	Visually checked during fitment of High Mount Stop Lamp	15 Out of 14 vehicles , associate not able to press 3 lugs uniformly	**Hypothesis Valid**

There if no reference for lug insertion



Uniformly pressing

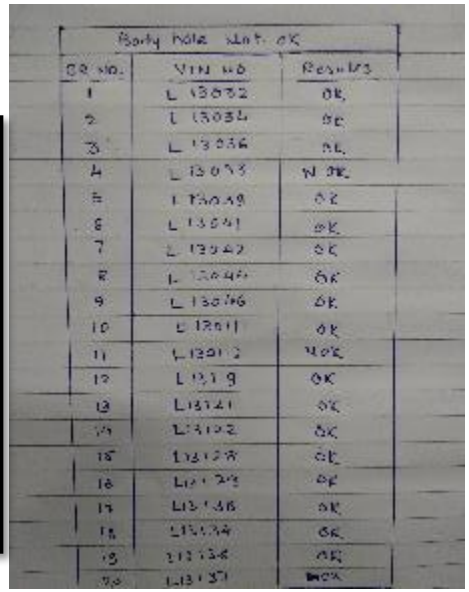
SR no	Results
1	NOK
2	NOK
3	NOK
4	NOK
5	NOK
6	NOK
7	NOK
8	OK
9	NOK
10	NOK
11	NOK
12	NOK
13	NOK
14	NOK
15	NOK

Inference- 15 vehicles checked & found associate not able to press High Mount Stop Lamp on veh.

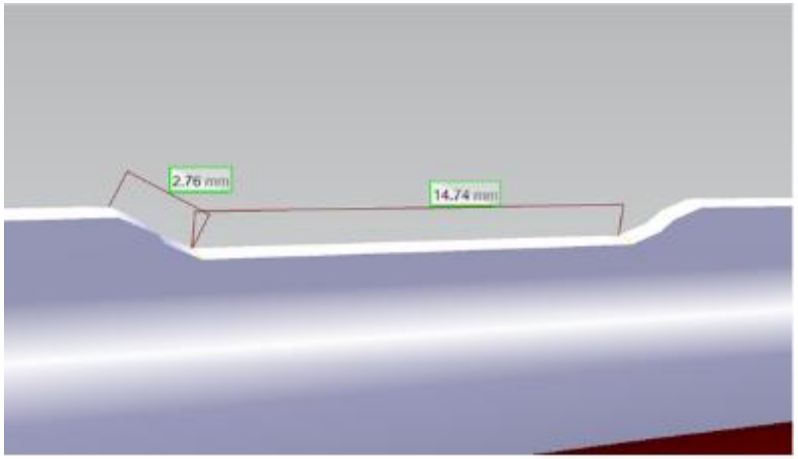
5.3 Confirm the probable cause:

No	Probable Cause	Testing method	Observations	Conclusion
6	Body hole not ok	Checked by ok High Mount Stop Lamp fitment on varies body hole.	Checked on stage with One High Mount Stop Lamp inserted on 20 nos. body holes , out of 18 nos. veh. Body hole found ok.	**Hypothesis Invalid**

Body hole not found ok



Checked by drawing



Inference- 20 No's Vehicles checked with High Mount Stop Lamp insertion ,out of this 18 vehicles fitment found ok, this is a invalid cause.

5.3 Confirm the probable cause:

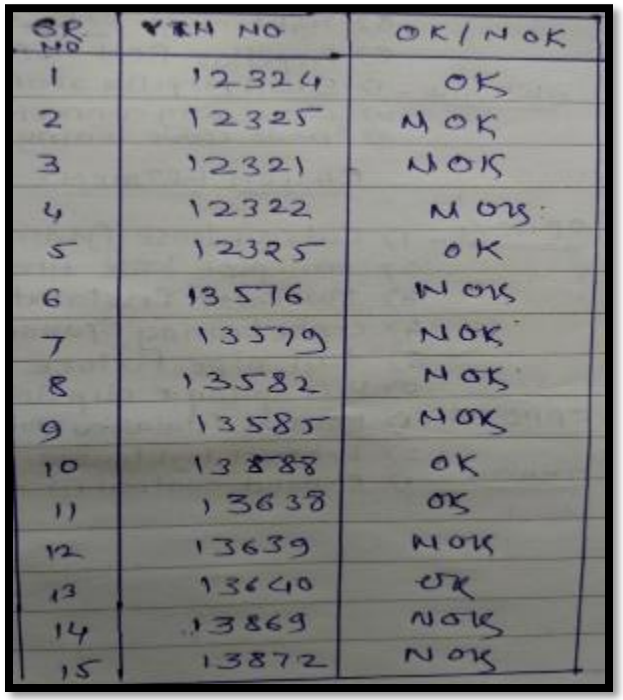
No	Probable Cause	Testing method	Observations	Conclusion
7	High Mount Stop Lamp lug poor pressing	Checked vehicles after fitment of High Mount Stop Lamp on body	15 vehicles checked and found associate not apply proper force during pressing	**Hypothesis valid**

Inference- 15 vehicles lug found poor locked , hence this is a valid cause.

5.3 Confirm the probable cause:

No	Probable Cause	Testing method	Observations	Conclusion
8	High Mount Stop Lamp lug over pressing.	Visually checked on stage	15 checked 10 found found not ok at CAI	**Hypothesis Valid**

Stage associate tapping on High Mount Stop Lamp during fitment



SR NO	VIN NO	OK/NOK
1	12324	OK
2	12325	NOK
3	12321	NOK
4	12322	NOK
5	12325	OK
6	13576	NOK
7	13579	NOK
8	13582	NOK
9	13585	NOK
10	13588	OK
11	13638	OK
12	13639	NOK
13	13640	OK
14	13869	NOK
15	13872	NOK

Inference –Vehicles found not ok ,this is a Valid Cause.

TEST OF HYPOTHESIS

5.2 Confirm the probable cause:

No	Probable Cause	Testing method	Conclusion
4	Push pin not locked	Visually	**Hypothesis valid**
6	Body hole not ok	Checked on stage with One HMSL inserted on 20 nos. body holes , out of 15 nos. veh. Body hole found ok	**Hypothesis Invalid**
8	High Mount Stop Lamp lug poor pressing.	Checked at CAI audit	**Hypothesis valid**
9	High Mount Stop Lamp lug over pressing.	Checked vehicles after fitment of High Mount Stop Lamp on body .	**Hypothesis valid**

5.3 Identify the root causes: (Use Why – Why)

WHY - WHY Analysis

Sr No	Valid Probable Cause	WHY	WHY	WHY	WHY	ACTION
6	Push pin not locked	Press three lug at a time	Three lug gap is more	No reference for installation	Blind operation	Ref marking to be provided on HMSL
7	HMSL lug poor pressing.	Lug not getting lock	Lug locking activity using hands	More effort required	All three lug press at a time	Uniformly pressing tool require
8	HMSL lug over pressing.	Lug not match with body hole	Insertion of lug on Body hole using hands	No confirmation of lug locking	Lug pressing with tapping on HMSL	Uniformly pressing tool require

6. Implement the corrective measures

Possible Remedies Identified

Legends

High ↑ ↓ Low

SN	Root Cause	Remedy Developed	Time to implement	Cost of Improvement	Easily reversible	Complexity	Decision
6	Blind operation	Ref marking to be provided on HMSL	Low	Medium	Medium	Medium	Accepted
7	All three lug press at a time	Uniformly pressing tool require	Low	Low	Low	Medium	Accepted
8	Lug pressing with tapping on HMSL	Uniformly pressing tool require	Low	Medium	High	Low	Accepted

6. Implement the corrective measures

6.1 Actions Decided

Root Cause	Action Plan	Target Dt	Resp.	Status
Blind operation	Reference marking provided on Tail side panel	20 Sep 18	Pravin Patil	Closed

Before

No marking for installation of HMSL lug in Respective location

After

Marking provided on body for reference on installation

6. Implement the corrective measures

6.2 Actions Decided

Root Cause	Action Plan	Target Dt	Resp.	Status
Uniformly pressing tool require	Fixture made for HMSL fitment	10 July 18	Pravin Patil	Closed.

Before

Before we fitment of HMSL with hand pressing

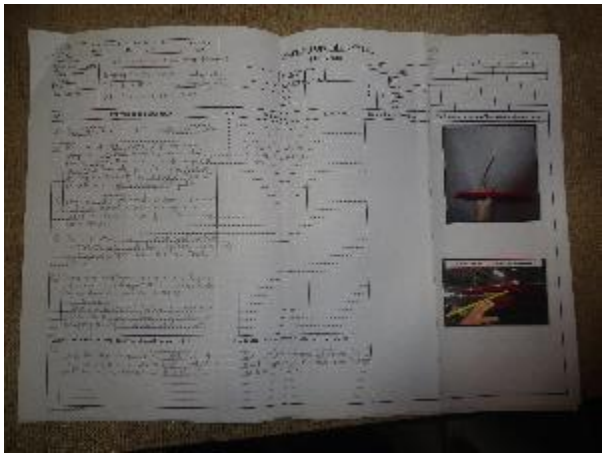
After

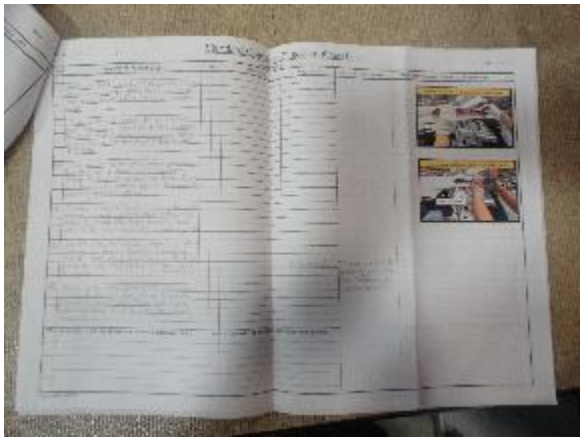
Uniformly pressing tool provided

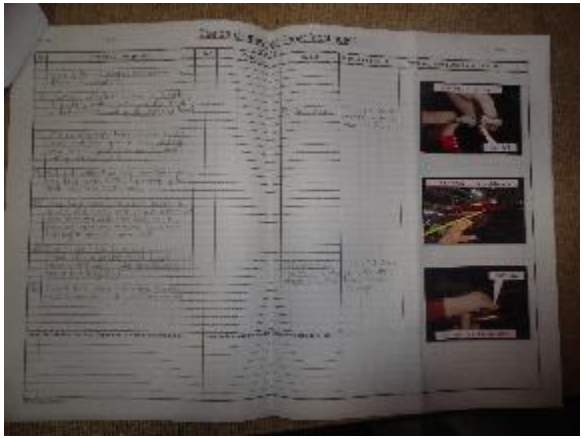
6. Implement the corrective measures

6.3 Actions Decided

High Mount Stop Lamp **fitment SOS :-**

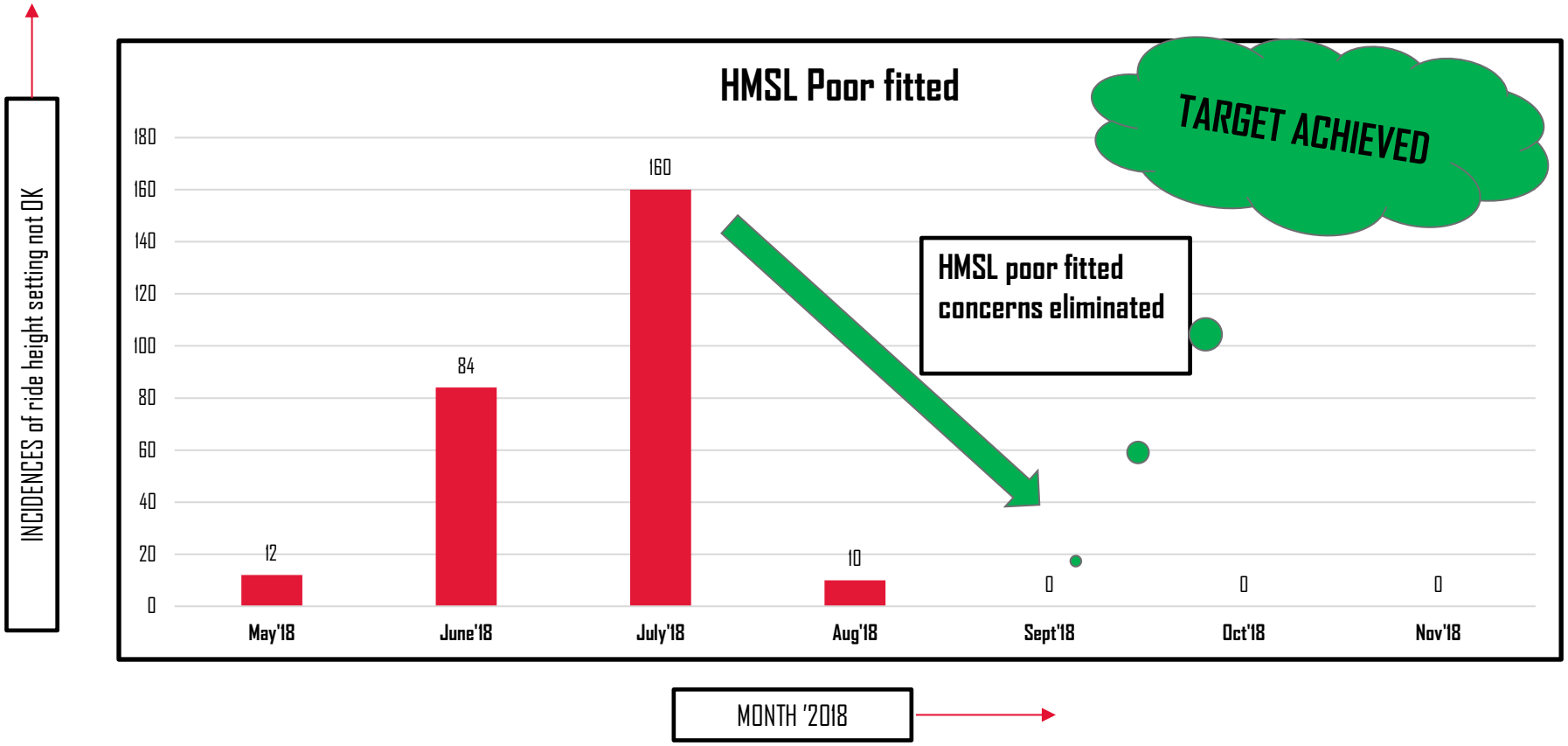






7. Check & confirm the results & effects

7.1 Verify Results:



Inference- After tacking PCA action, HMSL poor fitted concern eliminated.

Check & confirm the results & effects (Cont……)

7.3 Tangible benefits:

- REWORK cost saving – Rs. 52645 /Year.
- Avg. DPT of this concern was 255 is now reduced to Zero.

Wed 19-12-2018 19:04

DHANOKAR RAJESH

Re: MYB Cost Approval

To BHAMARE SACHIN

okay Approved

Sent from my Redmi Note 4X
On 19-Dec-2018 6:52 PM, BHAMARE SACHIN <BHAMARE.SACHIN@mahindra.com> wrote:
Sir,

Pl find the MYB project rework cost estimation for HMSL poor fit issue.

Project Leader – Pravin Patil.

Rework Cost estimation		
Description	Units	
Rework time (Approx)	Min	10
Avg Concerns/Day	Nos	6.0
Total Required Rework time per day	Min	60
Effective working time	Min	465
Operator required /day	nos	0.13
Approx Operator cost/ day	Rs	1360
Operator cost /concern	Rs	175.48
Avg no. of working days	days	25
Operator cost /month	Rs	4387.1
Operator Cost/year	Rs	52645.2

7.3 Intangible benefits:

1. Associate fatigue reduced fitment of HMSL on vehicle using fixture .
2. No brake leakage of water form HMSL so HMSL replace rework time eliminated.
3. FRC increased by 15 %.

8. Standardization

8.1 Updation of documents:

FMEA- HMSL fitment on body

Plant : Nashik	**Model :Marazzo**	***Mahindra & Mahindra Ltd.***			**Cross Functional Team**
Part Nos.:	**Part Name/ Process:** HMSL FITMENT				Sakharam Shelke, Dhikale
Latest Change Letter:	**Optn No.:** 1703003	***Process Potenial Failure Mode & Effect Analysis.***			Vikrant Ghule, Santosh Jadhav
PU/ Shop :Scorpio TCF	**Optn Desptn:** HMSL FITMENT	**Reviewed by :Sakharam Shelke**	**Rev. No :**	**Doc No.: MQS/PFMEA/NASHIK/U321/TCF/ 1703003**	Sayali More
Module/ Cell : Trim 03	**M/C Name & No.:**	**Approved by : Mandar Joshi**	**Rev. Date :**	**FMEA Date (Orig) :18-07-20**	

Process Function	Requirements	Potential Failure Mode	Potential Effect(s) of Failures	sev	class	Potential Cause(s)/Mechanism(s) of Failures	Current Process Controls Prevention	Occur	Current Process Controls Detection	Detect	RPN	Recommended Action(s)	Responsibility & Target Completion Date	Action Results: Action Taken	Sev	Occ	Det	RPN
TAKE HMSL AND CONNECT CONNECTOR OF LIFT GATE HARNESS TO HIGH MOUNT STOP LAMP	PROPER CONNECTION	IMPROPER INSERTION	**Mfg. & Assy:** REWORK **End User:** CUSTOMER DISSATISFACTION	6	SC	OPERATOR MISTAKE	AS PER PROCESS SHEET & SOP	5	ECOS CHECK	7	210	ECOS CHECK	CME	POINT ADDED IN ECOS CHECKING	6	3	5	90
PULL REAR WASHER TUBE FROM LIFT GATE CUTOUT AND CONNECT TO HMSL	PROPER CONECTION	LOOSE CONNECTION	**Mfg. & Assy:** REWORK **End User:** CUSTOMER DISSATISFACTION	4		OPERATOR MISTAKE	OPERATOR TRAINING	3	SELF CHECK	7	84							
HIGH MOUNT STOP LAMP & CLIP IT ON TAILGATE OUTER PANEL (M)	PROPER FITMENT	IMPROPER INSERTION	**Mfg. & Assy:** REWORK **End User:** CUSTOMER DISSATISFACTION	4		OPERATOR MISTAKE	AS PER PROCESS SHEET & SOP	3	TRIM BUYOFF CHECK	7	84	HMSL fitment using Fixture	Pravin Patil					

Rev No.	Rev Date	Revision Summary	Reason for Revision
01	18-07-2017	New Release	FMEA made for Marazzo vehicle
2	18-08-2017	Fixture activity added	For concern of HMSL POOR fitted

8. Standardization

8.1 Updation of documents:

PCRN- HMSL fitment on body using fixture .

AY PCRN

PCRN/A003/STCF/19/12047 | Introduction of Marazzo Vehicle | Planned closing date: * 30.10.2018 | Actual closing date:

STCF | Cell: * ALL | Date: 17.10.2018

Marazzo

Opn Description: Introduction of Marazzo Vehicle

e Trigger

ameters

- [ ] Layout Change
- [ ] Use of alternate material
- [ ] Product produced after the tooling has been inactive for more than 12 month
- [ ] Production from New M/c
- [ ] 25% prodn. Ramp UP than rate quoted in present PPAP
- [ ] Change of Vendor for Part, Material, Paint etc
- [ ] Poka Yoka Introduction
- [ ] Transferred M/c
- [ ] Production from tooling and equipment transferred to a different plant location or from additional plant location Production from New or Modified
- [ ] Inspection & Test Method
- [ ] Any Other (Kaizens/ Improvement Projects/ Ingenious Idea etc.)
- [ ] Reconditioned M/c/ Toolir

Marazzo Vehicle

Impact

Impact: ○ Saving ○ Increase ◉ Others

[ ] Manpower [ ] Conversion

ing PU/ Area / Shop

cuments(Attachments limit is 1mb/PCRN no.)

ly Implemented: * ○ Yes ◉ No Implementation Date: *

ollowed: * ◉ Yes ○ No SHE approval required ◉ Yes ○ No

ssary from (if affected)

upply Module ◉ Yes ○ No ii.Maintenance Department ◉ Yes ○ No iii.Quality Assurance ◉ Yes ○ No

itatus

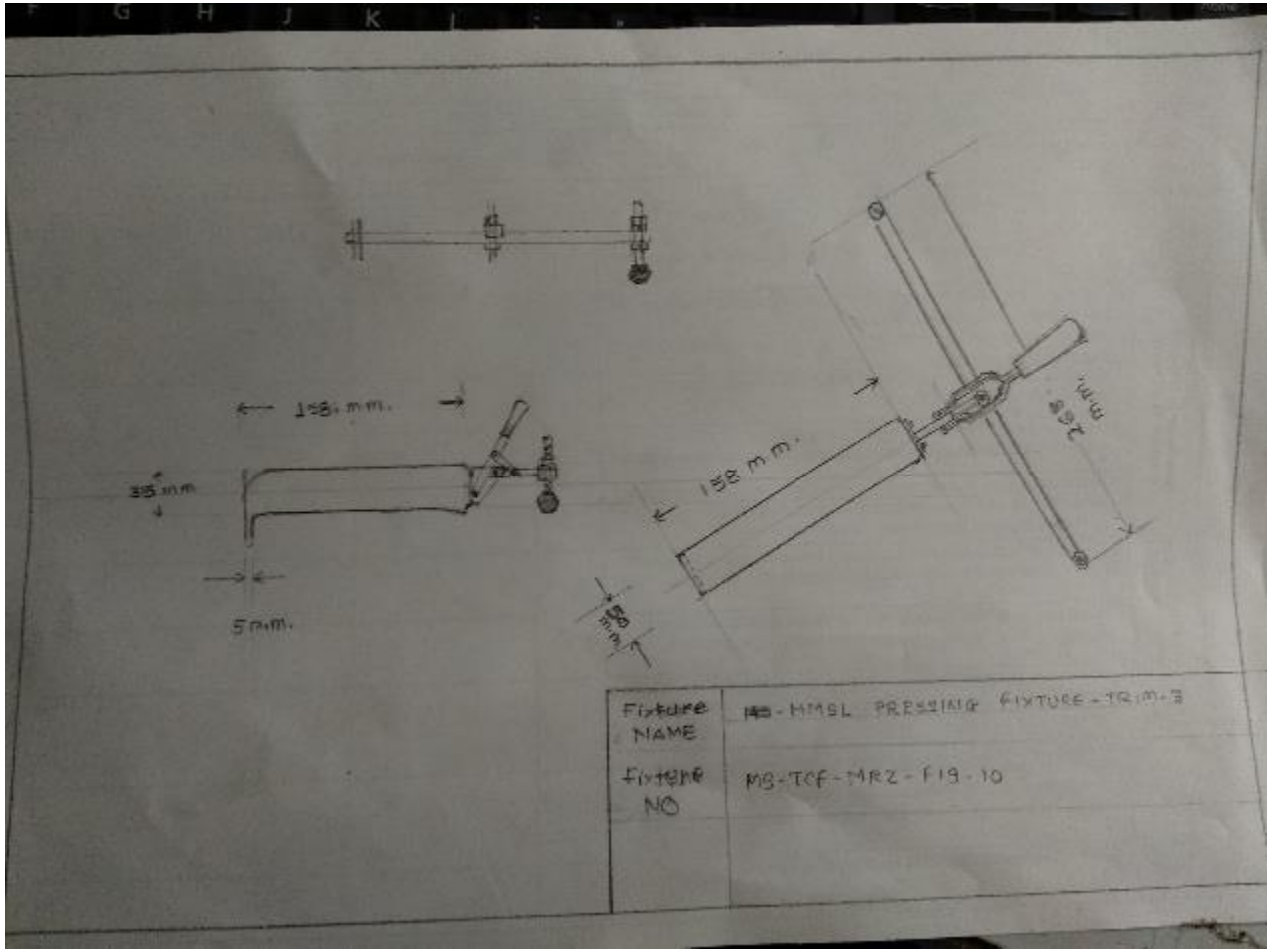
itus

nce	SCM	Quality	PU Head	SHE Head	Head Plant Quality	Plant Engg Co-Ordinator	Plant Engg Manager	Plant Engg Head
	■	■	■	■	△	△	△	△

8. Standardization

8.1 Updation of documents:

Fixture Drg. Updated



8. Standardization

8.2 Training :-

Training given to all stage associate for HMSL fitment with fixture

8. Standardization

8.3 Updation of documents:

Process audit sheet modified for HMSL fitment using fixture

Marazzo - Process Audit check sheet (Trim 3 Line)				Date →											Month -			
Sr No	BWT	Stage / Fitment process	What to check	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
1	4	HMSL fitment	Use of fixture for HMSL Fitment															
2	4	Jack Assly	Strap tightly assembled & cross marking done on it.															
3	4	All pillar / RQT	No Scratches on part due to handling / storage															
4	5	2nd row seat fitment	1. Tightening sequence followed at 2nd row seat mounting bolt 2.Torque application all bolts (at further															
5	5	Water shield fitment	Roller application on sealant & marking – no wrinkle on water shield															
6	5	Trim Pad fitment	1. Ensure power window switch connection – Socket fully locked. 2.Ensure no gap and fitment sequnce for power window switch bezel 2. Trim Pad locators correctly aligned while fitment on Door panal. 3. All clips fully locked on body panal. 4.Patch (power window sw.) locking procedure															
7	5	Speaker fitment	Use of special gun for speaker fitment 3Nm															
8	5	Qtr. Window outer waist seal fitment	1. promoter application for tailgate garnish & at qtr window outer waist seal fitment with use of fixture. 2. outer waist seal press firmly by hand at its															
9	5	Outer waist seal fitment	1. Ensure no use of mallet during outer waist seal fitment 2 Waist seal pasted by hand for chrome															
10	6	Tailgate garnish fitment	primer application															
11	6	Windsield Fitment	1. Use of 3 fixture for glass for glass alignment. (2 at top side & 1 at LH A pillar) 2. Ensure glass is pressed completely at all of its edges. 3. sealer height & width checked by S.O. with scale & updated on graph.															
12	6	Tailgate glass fitment	1. positioning of spacers															
13																		
SPU/TCF/MQA/F/09														Rev.- 03 effective from 15.07.18				

9. Summaries & Plan future actions

9.1 Conclusion:

High mount top lamp poor fitment issue eliminated after Implementing the pressing fixture.

9.2 Horizontal Deployment:

All U322 and U325 model From VP gateway.

9.3 Lessons Learnt:

Things Gone Wrong :

Initially all focus given to increase the Press panal slot width increase**.**

Things Gone Right :

Instead of going with Design change team implemented small inhouse developed fixture which presses at exact clip locations & reduces the efforts & all 3 clips gets locked at a time that means uniformly it gets pressed.

Thank you……!